1. Microorganism of plant food products.
2. Microorganism of Animal food products.
3. Microorganism of processed food.
4. Standard chemicals added to the food.
5. Excess use of preservatives in food.
6. Presence of higher population of Microorganism in food.
7. Toxin produced by various types of Microorganism.

## 5.4 Food Preservation Methods

Foods can be preserved by a variety of methods. It is vital to eliminate or reduce the populations of spoilage and disease – causing microorganisms and to maintain the microbiological quality of a food with proper storage and packaging. Contamination often occurs after a package or can is opened and just before the food is served. This can proved an ideal opportunity for growth and transmission of pathogens, if care is not taken. Preservation of food is the process by which food is stored by special methods. Cooked or uncooked food can be preserved in different ways to be used later Table 5.2. Some methods of preservation are:

### 1. Freezing

Food kept in a refrigerator remains fresh for some day. Germs do not grow easily in cool places we preserve food items, like milk, fruit, vegetable and cooked food by keeping them in a refrigerator.

### 2. Boiling

By this method, we can preserve food for a short period of time. Germs in milk are killed by pasteurization. It is done by boiling milk for sometimes and then cooling it quickly.

### 3. Salting

Add salt to preserve pickles and fish.

### 4. Sweetening

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Sugar act as a preservative when added in large quantities. For example, food can be stored for a long time in the form of jams, jellies and murabbas (Figure 5.2) by adding sugar.



**Figure 5.2:** Murabbas

### 5. Drying

In this method, the food items are dried in sun to stop the growth of bacteria in them. Certain foods, like raw mangoes, fishes, potato chips and papads are preserved by this method.

### 6. Canning

In this method, air is removed from food and put in airtight cans so that germs do not grow on them. Food items like vegetables, seafood, and dairy product are preserved through this method.

### Advantages of food preservation

- Germs do not grow easily in preserved food and make it safe to eat.
- Preservation enables us to enjoy seasonal fruits like strawberries and mangoes even during the off season.



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